

EE3005 Communication Systems - Assignment 1

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Topics covered: Fourier transforms, basic communication channels, signal representation, up/down-conversion, heterodyne.

Instructions: Submit your solutions in one PDF file with all step-by-step answers, plots and outputs written/printed in that file.

1. A system with input $x(t)$ has output given by

$$y(t) = \int_{-\infty}^t e^{u-t} x(u) du$$

- Show that the system is LTI and find its impulse response.
 - Find the transfer function $H(f)$ and $|H(f)|$.
 - Let the input $x(t) = 2 \text{sinc}(2t)$, find the energy of the output.
2. Use Parseval's identity to compute the following integrals.
- $\int_{-\infty}^{\infty} \text{sinc}^2(2t) dt$
 - $\int_0^{\infty} \text{sinc}(t) \text{sinc}(2t) dt$
3. Derive the Fourier Transform of the signal $g(t) = \frac{1}{\pi t}$ using the Duality Principle. This operation is known as the Hilbert Transform. Also, determine the specific phase shift it imposes on positive and negative frequencies. *Hint: Use the duality property starting with the transform pair $\text{sgn}(t) \longleftrightarrow \frac{2}{j\omega}$.*
4. The system shown in Figure 1 is used to scramble audio signals. The output $y(t)$ is the scrambled version of the input $m(t)$

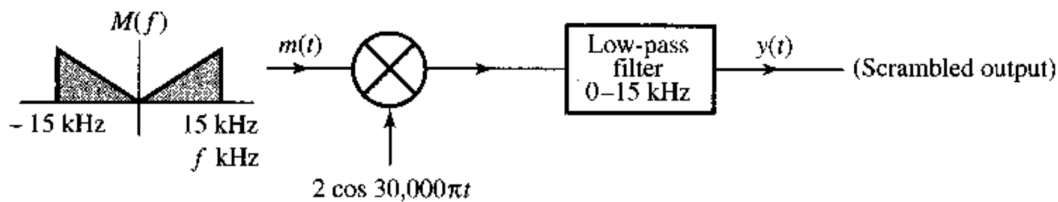


Figure 1: Audio signal scrambling system

1. Find the spectrum of the scrambled signal $y(t)$
2. Suggest a method of descrambling $y(t)$ to obtain $m(t)$
5. Let $x(t)$ and $y(t)$ denote two bandpass signals, and let $x_l(t)$ and $y_l(t)$ denote their lowpass equivalents with respect to some frequency f_0 . We know that in general $x_l(t)$ and $y_l(t)$ are complex signals.

a) Show that

$$\int_{-\infty}^{\infty} x(t)y(t) dt = \frac{1}{2} \Re \left[\int_{-\infty}^{\infty} x_l(t)y_l^*(t) dt \right]$$

b) Also show the energy of bandpass signal is one-half of the energy of its lowpass equivalent

6. A real passband signal is given by

$$x(t) = 6 \cos \left(2\pi f_c t + \frac{\pi}{6} \right) - 4 \sin \left(2\pi f_c t + \frac{\pi}{6} \right)$$

a) Express $x(t)$ in the form

$$x(t) = I \cos(2\pi f_c t) - Q \sin(2\pi f_c t)$$

and find I and Q .

- b) Find the complex baseband equivalent signal $x_b(t)$.
- c) Compute the envelope and phase of the signal.
7. Suppose that a passband signal $u_p(t)$ is passed through a passband filter with impulse response $h_p(t)$ to produce an output $y_p(t)$. Let $u(t)$, $h(t)$, and $y(t)$ denote the corresponding complex envelopes of $u_p(t)$, $h_p(t)$, and $y_p(t)$, respectively, with respect to a common reference frequency f_c . Show that the complex envelope of the output satisfies

$$y(t) = \frac{1}{2} (u * h)(t)$$

where $*$ denotes convolution.

8. Let $u_c(t)$ and $u_s(t)$ be two baseband signals and let $u_p(t)$ be passband signal.
 - a) Can we get $u_c(t)$ and $u_s(t)$ back from $u_p(t)$?
 - b) Can any real passband signal be represented in this form?
9. Let $v_p(t)$ denote a real-valued passband signal with Fourier transform $V_p(f)$ specified as follows for negative frequencies:

$$V_p(f) = \begin{cases} -(f + 99), & -101 \leq f \leq -99 \\ 0, & f < -101 \text{ or } -99 < f \leq 0 \end{cases}$$

- a) Sketch $V_p(f)$ for both positive and negative frequencies.
- b) Find and sketch $V_c(f)$ and $V_s(f)$, the Fourier transforms of the I and Q components with respect to a reference frequency $f_c = 99$. Do this without going to the time domain.
- c) Find an explicit time-domain expression for the output when $v_p(t) \cos(200\pi t)$ is passed through an ideal lowpass filter of bandwidth 4.

10. The passband signal

$$u(t) = \mathbf{1}_{[-1,1]}(t) \cos(100\pi t)$$

is passed through the passband filter

$$h(t) = \mathbf{1}_{[0,3]}(t) \sin(100\pi t).$$

Find an explicit time-domain expression for the filter output. (Hint: use complex envelopes convolution property)

11. A communication signal occupies the frequency band

$$[250, 270] \text{ MHz}$$

The signal is downconverted using a heterodyne receiver with local oscillator frequency

$$f_{LO} = 300 \text{ MHz}.$$

- a) Find the frequency bands present at the output of the mixer.
- b) Identify the intermediate frequency (IF) band.
- c) Compute the IF bandwidth.

12. A received signal is given by

$$r(t) = 5 \cos(2\pi f_c t + \theta).$$

The signal is demodulated using a homodyne receiver.

- a) Find the outputs of the in-phase (I) and quadrature (Q) branches after low-pass filtering.
- b) Show how the phase θ can be recovered from the I and Q components.

13. A communication system operates over an AWGN channel with bandwidth $B = 3 \text{ MHz}$ and noise PSD $N_0 = 5 \times 10^{-10} \text{ W/Hz}$. If the signal power is $P_s = 15 \text{ mW}$, calculate:

- a) SNR in dB
- b) Maximum bandwidth if the required SNR is 15 dB

14. A signal

$$s(t) = \sqrt{2E} \cos(2\pi f_c t)$$

is transmitted over an AWGN channel with two-sided noise power spectral density $N_0/2$.

- a) Write the expression for the received signal $r(t)$.
 - b) The received signal is passed through a coherent receiver matched to $s(t)$. Derive the expression for the signal-to-noise ratio (SNR) at the output of the matched filter.
 - c) If the received signal energy is reduced by a factor of 4 due to channel attenuation, determine the new output SNR in terms of E and N_0 .
 - d) Find the ratio of output SNRs before and after attenuation.
15. If a signal $s(t)$ is corrupted by AWGN, show that the filter with impulse response matched to $s(t)$ maximizes the output SNR.

16. Consider a baseband channel with impulse response

$$h(t) = \delta(t) - 0.8\delta(t - T) + 0.3\delta(t - 2T),$$

where T is the symbol period.

The coherence bandwidth is defined as

$$B_c \approx \frac{1}{\tau_{\max}},$$

where $\tau_{\max}$ is the maximum excess delay.

- a) Compute the coherence bandwidth B_c .
 - b) Find the frequency response $H(f)$.
 - c) Determine whether a system transmitting at symbol rate $\frac{1}{T}$ will experience ISI. Justify briefly.
17. A communication system transmits binary data at a rate of $R_b = 9600$ bits/s using Pulse Amplitude Modulation (PAM).
- a) According to Nyquist's First Criterion, what is the absolute minimum theoretical bandwidth required to transmit this signal with zero ISI?
 - b) If the channel bandwidth is limited to 4 kHz, can this bit rate be achieved using binary signaling without ISI? Justify your answer.
18. The Fourier transform $P(\omega)$ of a basic pulse $p(t)$ used in certain binary communication system is shown in Figure 5

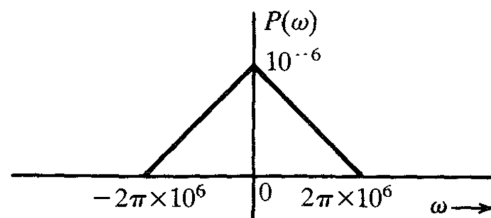


Figure 5: Spectrum $P(\omega)$

- a) From the shape of $P(\omega)$ explain at what pulse rate this pulse would satisfy the Nyquist's criterion.
- b) Find $p(t)$ and verify that this pulse does (or does not) satisfy Nyquist's criterion. If it does satisfy the Nyquist's criterion what is the transmission rate in bits per sec?
- c) What is the roll-off factor?